WORKSHEET 6.9

Integration by Substitution

Unit 6: Integration and Accumulation of Change

Key ideas

- Substitution undoes the chain rule. Choose u to be an **inside** function whose derivative appears as a factor.
- Replace every x and the dx: if $u = g(x)$ then $du = g'(x)\,dx$.
- A missing constant factor can be supplied: if $du = 2x\,dx$, then $x\,dx = \frac{1}{2}du$.
- In a definite integral, **change the limits** to values of u, or return to x before evaluating. Never mix the two.

Worked examples

Example 1 · A direct fit

Find $\displaystyle\int 3x^2\left(x^3+5\right)^4\,dx$.

Solution. $u = x^3+5$, $du = 3x^2\,dx$: $\displaystyle\int u^4\,du$.

$\dfrac{(x^3+5)^5}{5} + C$.

Example 2 · A linear inside

Find $\displaystyle\int \cos(4x)\,dx$.

Solution. $u = 4x$, $dx = \frac{1}{4}du$.

$\dfrac{1}{4}\sin(4x) + C$.

Example 3 · Changing the limits

Evaluate $\displaystyle\int_0^3 \frac{2x}{x^2+1}\,dx$.

Solution. $u = x^2+1$, $du = 2x\,dx$; when $x=0$, $u=1$; when $x=3$, $u=10$. $\displaystyle\int_1^{10}\frac{1}{u}\,du$.

$\ln 10$.

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Practice

CORE

1. Find $\int 2x\left(x^2+1\right)^3\,dx$.

2. Find $\int e^{5x}\,dx$.

3. Find $\int \sin(3x)\,dx$.

4. Find $\int \dfrac{1}{2x+1}\,dx$.

EXAM LEVEL

5. Find $\int x\sqrt{x^2+9}\,dx$.

6. Evaluate $\displaystyle\int_0^{\pi/2} \sin^2 x\cos x\,dx$.

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7. Evaluate $\int_0^1 x^2 \left(x^3+1\right)^3 \, dx$.

8. Find $\int \cot x \, dx$ by writing $\cot x = \dfrac{\cos x}{\sin x}$.

Question 9 · Free-response practice

Answer each part. No calculator is permitted.

(a) Evaluate $\int_0^2 \dfrac{x}{x^2+4} \, dx$.

(b) Evaluate $\int_0^\pi \sin x \, e^{\cos x} \, dx$.

(c) If $\int_0^8 f(x) \, dx = 20$, find $\int_0^2 f(4x) \, dx$.

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Worked solutions

1. $u = x^2 + 1$: $\dfrac{(x^2+1)^4}{4} + C$.

2. $\dfrac{1}{5}e^{5x} + C$.

3. $-\dfrac{1}{3}\cos(3x) + C$.

4. $\dfrac{1}{2}\ln|2x+1| + C$.

5. $u = x^2 + 9$: $\dfrac{1}{2} \cdot \dfrac{2}{3}u^{3/2} = \dfrac{1}{3}(x^2+9)^{3/2} + C$.

6. $u = \sin x$ from 0 to 1: $\left[\dfrac{u^3}{3}\right]_0^1 = \dfrac{1}{3}$.

7. $u = x^3 + 1$ from 1 to 2: $\dfrac{1}{3}\left[\dfrac{u^4}{4}\right]_1^2 = \dfrac{15}{12} = \dfrac{5}{4}$.

8. $u = \sin x$, $du = \cos x\,dx$: $\ln|\sin x| + C$.

9. (a) $u = x^2 + 4$ from 4 to 8: $\dfrac{1}{2}\Big[\ln u\Big]_4^8 = \dfrac{1}{2}\ln 2$.

   (b) $u = \cos x$ from 1 to -1, $du = -\sin x\,dx$: $\displaystyle\int_{-1}^{1} e^u\,du = e - \frac{1}{e}$.

   (c) $u = 4x$, $dx = \dfrac{1}{4}du$, limits 0 to 8: $\dfrac{1}{4}(20) = 5$.

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